

**Ques. WHAT IS DEEP LEARNING?****Ishita Bhatnagar**

Ans. Deep learning allows computers to learn on their own, with some guidance. It is machine learning that tries to mimic the way a human brain works, to get closer to the real meaning of artificial intelligence (AI). Machine learning is about teaching a machine to do something. This technology uses a combination of feature extraction and modality-specific machine learning algorithms, along with thousands of examples, to teach a machine to identify things like handwriting and speech.

The process is not as easy as it sounds. It requires a large set of data, heavy computing power and a lot of background work. And, despite tedious efforts, such systems are not fool-proof. Deep learning tries to solve these problems and take machine learning one step ahead.

A deep learning computer learns by itself. It is made up of multilayered deep neural networks that mimic the activities of the layers of neurons in the neo-cortex part of human brain. Deep learning uses large number of processors and equally huge amounts of data to feed to those processors.

You may refer to 'Uncanny Vision Uses Deep Learning To Sense Unfamiliar Happenings' and 'Deep Learning Makes Conventional Machine Learning Look Dumb' articles published in January issue.

Q2. HOW DOES AN INKJET PRINTER WORK?**Himanshu S.**

A2. An ink-filled print cartridge attached to the inkjet's print-head moves

sideways across the width of a sheet of paper that is fed through the printer below the print head. The print-head usually contains four ink cartridges—one each for magenta (red), cyan (blue), yellow and black. Each cartridge is made up of around 50 ink-filled firing chambers, each attached to a nozzle smaller than a human hair.

An electrical pulse flows through thin resistors at the bottom of all chambers of all colours that the printer uses to form a small section of a character or picture on paper. When an electrical current flows through a resistor, it heats a thin layer of ink at the bottom of the chamber to more than 482°C for several millionths of a second. The ink boils and forms a bubble of vapour.

As the vapour bubble expands, it pushes ink through the nozzle to form a droplet at the tip of a nozzle. The droplet overcomes the surface tension of ink, and pressure of the vapour bubble forces the droplet onto the paper. Volume of the ejected ink is about one-millionth of a drop of water from an eyedropper. A typical character is formed by an array of these drops, 20 across and 20 high.

As the resistor cools, the bubble collapses. The resulting suction pulls fresh ink from the attached reservoir into the firing chamber.

Q3. HOW DO DIGITAL SUBSCRIBER LINES, OR xDSLs, WORK?**N. Tiwari**

A3. There are several forms of digital subscriber lines, or xDSLs, with x depending on a particular variety of DSL. All xDSL connections use the same ordinary pair of twisted copper wires that already carry phone calls

among homes and businesses.

Unlike cable modem connections, which broadcast everyone's cable signals to everyone on a cable hub, xDSLs are point-to-point connections, unshared with others using the service.

Signals travel between a network interface in your PC and an xDSL modem. You do not have to dial up an Internet service provider; your net connection is always on. Most modems have proprietary designs that require the local phone company to use specific equipment.

You can use the same phone line for the Internet service at the same time it is carrying a voice call because the two signals use widely-separated areas of the frequency spectrum. A splitter next to your xDSL modem combines low-frequency voice signals and higher-frequency data signals.

The most common form of xDSL is ADSL, where A stands for asynchronous, meaning that, more bandwidth or data-carrying capacity is devoted to data travelling downstream from the Internet to your PC as compared to upstream data travelling from your PC to the Internet. The reason for the imbalance is that, upstream traffic tends to be limited to a few words at a time.

Transmission rates depend on the quality of the phone line, type of equipment it uses, distance from the PC to a phone company switching office and type of xDSL being used. A splitter on the other end of the line breaks voice and data signals apart again, sending voice calls into the plain old telephone system and computer data through high-speed lines to the Internet.

Answers compiled by EFY senior application engineer, Nidhi Kathuria. Letters and questions for publication may be addressed to Editor, Electronics For You, D-97/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editor@efy.in) and should include name and address of the sender